

Question 1

Consider country Alpha, where the current and the future expected inflation rates are both zero, so that real and nominal interest rates are equal. In period $t + 1$ ("the future"), consumption depends just on disposable income at $t + 1$, and investment on the interest rate and output at $t + 1$ only. Turning now to the current period t , the present, consumption is increasing in both current and expected future disposable income, $C_t = C(Y_t - T_t, Y_{t+1}^e - T_{t+1}^e)$; investment is increasing in current and future expected income, and decreasing in the current and future expected short-term interest rates; finally, G and T are exogenous as usual. The IMF today (time t) publishes new statistics revising downwards the estimated demand for the future.

Suppose that only one-year bonds and two-year bonds exist and that, before the announcements described above, at time t the yield curve was upward sloping. Moreover, assume that the central bank announces today that the bank will enact any policy that would result to be necessary in the future to keep constant the level of GDP. What will be the effects of the new forecasts and the CB's policy announcement on the position and the slope of the yield curve of the country? Explain your answer.

Solutions

Given the prediction of the IMF, agents expect lower demand in the future: all else equal, the future expected IS will shift left, resulting in lower expected income. If the CB is credible, markets expect that in $t + 1$ an expansionary monetary policy will be conducted, reducing i_{t+1}^e . In the current period, the Central Bank is not changing the interest rate, hence i_{1t} is constant.

We can conclude that i_{1t} does not change, while $i_{2t} \approx \frac{i_t + i_{t+1}^e}{2}$ declines.

The yield curve rotates downward: we know for sure that it will be flatter than before, but we can't know whether it will be still upward sloping, flat or even downward sloping.

Question 2

Consider an economy, Macrolandia, freely trading goods, services and financial assets with the rest of the world (Foreign country), in a flexible exchange rate regime. Domestic and foreign prices are constant, and equal to one ($P = P^* = 1$).

Using the pair of graphs 'open economy IS-LM - interest parity condition', suppose now that the foreign country adopts a policy mix such that the foreign interest rate, i^* , increases, while foreign GDP, Y^* , remains constant.

Describe the impact of the policy change in the foreign country on the economy of Macrolandia and provide the economic intuition.

Moreover, assume that before the shock Macrolandia's trade were balanced ($NX = 0$). What happens to net exports after the shock?

Solutions:

If $i^* \uparrow$, foreign assets, all else equal, become more convenient and, given the expectations about the future exchange rate, the euro depreciates: the UIP line rotates inwards. (Point A , always belonging to the UIP , shifts up). The exchange rate declines, stimulating net exports (if Marshall-Lerner conditions hold), increasing demand and production: the IS shifts to the right. The impact is an increase in GDP and a depreciation of the euro against the foreign currency, while the domestic interest rate is constant.

There are two effects on net exports: the depreciation of the exchange rate leads to an improvement in the trade balance, while an increase in equilibrium GDP, increasing imports, deteriorates net exports.

However, the impact is not ambiguous, as we can understand using the graph of the goods market. The depreciation of the euro shifts up both the NX line and the Z (demand) line, while D , demand in the closed economy is not affected. Given that Z is flatter than D , this means that point 2, the new equilibrium, lies at the left of the new intersection between Z and D after the shock, denoting the trade balance GDP level.

Question 3

True or False?

Explain whether the following statements are true or false. Motivate your answer in a brief but rigorous way, by making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

Consider a country that, in period $t = 1$ inherits from the previous period $t = 0$ a debt-to-GDP ratio of 100%: $b_0 \equiv B_0/Y_0 = 1$. Moreover, the real interest rate is constant and equal to 2%. Finally, the primary deficit to GDP ratio is 3%, assumed to be constant over time. In order to keep stable the level of debt to GDP over time, the growth rate of the economy should be at least 4%.

Solutions:

False!

The equation describing the evolution of the debt-to-GDP ratio is:

$$b_t = (1 + r - g)b_{t-1} + d_t$$

Assume the ratio of debt-to-GDP is constant to $\bar{b}$: $\bar{b} = (1 + r - g)\bar{b} + d_t \Rightarrow \bar{b}(g - r) = d_t$.
To find the growth rate such that debt is constant at this value, we solve for g : $bg = d_t + r\bar{b}$

$$g = d/\bar{b} + r = 0.03/1 + 0.02 = 0.05$$

Hence, the statement is **false**. The growth rate must be 5%. In fact, if the growth rate were 4%,

$$b_1 = (1 + 0.02 - 0.04)b_0 + 0.03 = 0.98 \times 1 + 0.03 = 1.01 > 1$$

While, with $g = 5\%$:

$$b_1 = (1 + 0.02 - 0.05) \times 1 + 0.03 = 0.97 + 0.03 = 1$$

Question 4

True or False?

Explain whether the following statements are true or false. Motivate your answer in a brief but rigorous way, by making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

The one-year interest rate in the US is 2%, the domestic interest rate in the euro area is 1% and the nominal exchange rate between the US dollar and the euro is 1.1 (the number of dollars to purchase a euro). Assuming the financial markets are perfectly operating, i.e. the interest parity condition holds, this means that markets expect an appreciation of the euro in the future.

Solutions:

TRUE

$i^* = 2\%$; $i = 1\%$, and $E = 1.1$ dollars per euro.

The interest parity condition (approximate version is):

$$i = i^* - \frac{E^e - E}{E}$$

Rearranging to solve for E^e :

$$\frac{E^e - E}{E} = i^* - i = 1\%$$

Markets expect an appreciation of the euro in the future of exactly 1%, that is the difference between interest rates paid by foreign assets and domestic assets.

In fact, in order to have both types of assets traded on the market, it must be that, the expected return on the two bonds is exactly the same. If US bonds pays higher rates in dollar terms, this means that a depreciation of the dollar is expected in the future to make the expected return in euros on those assets equal to the return people would get buying the domestic bond.